MIL-R-4286E

22 June 1972
SUPERSEDING
MIL-R-4286D
30 September 1970

MILITARY SPECIFICATION

* RIVETERS, PNEUMATIC, COMPRESSION TYPE, PORTABLE, WITH RIVET SETS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers portable, compression-type, pneumatic riveters with rivet sets.

1.2 Classification.- Riveters covered by this specification shall be of the following types and classes, as specified (see 6.2).

* Type I - Alligator jaw type

Class 1 - 1-1/2 inch reach
Class 1L - 9-1/8 inch reach
Class 2 - 5 inch reach
Class 3 - 2-7/8 inch reach

Type II - C - Yoke compression riveters

Class 1 - 1-1/2 inch reach
Class 3 - 2-1/8 inch reach

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-P-40 - Packaging and Packing of Hand Tools

FSC 5130

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MS - 20470	-	Rivet, Solid, Universal Head, Aluminum and Aluminum Alloy
MS - 20613	-	Rivet, Solid-Universal Head, Steel, Carbon and Steel Corrosion Resistant
MS - 20615	-	Rivet, Solid-Universal Head, Brass Copper, and Nickel Copper Alloy

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook H-28 - Screw - Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Standard product.- The riveters covered by this specification shall be the current standard product of the manufacturer, except for changes necessary to comply with this specification.

3.2 Materials.- Material not definitely specified shall be of the quality normally used by the manufacturer for riveters, provided the completed item complies with all provisions of this specification.

* 3.3 Design and construction.- The riveters shall be a manually actuated, pneumatically operated tool, designed for operation on an air pressure of 90 to 100 pounds per square inch gage (psig). The riveters shall consist of a cylinder, a set of jaws, a throttle and a handle. The riveters shall accommodate rivet sets conforming to 3.5. The riveters shall be designed to safely contain and exert the riveting pressures listed in tables I and II for the sizes furnished. When tested as specified in 4.3.1 and 4.3.2, there shall be no failure of operation, fracture, deformation, loosening of parts, damage to the riveter, riveter jaws or rivet sets.

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3.3.1 Throttles.- Each type I and type II riveter shall be provided with a nonleakable throttle of the lever type for controlling the air supply and for regulating the speed of the riveter.

3.3.2 Handles.- Riveters with overall lengths of 17 inches or more shall be provided with straight handles attached to the end of the riveter opposite the jaws. Riveters less than 17 inches in length shall be of such form that hand grips can be secured on the cylinder housing.

3.3.3 Air inlet connections.- The air inlet connections to the riveters shall be provided with 1/4 inch internal pipe thread. The air inlet shall have a screen to exclude any particles from entering air controlled valves.

3.3.4 Welding, brazing and soldering.- Welding, brazing, and soldering shall be performed only on those operations included in the original fabrication of the riveters. Welding, brazing and soldering shall not be utilized as a repair measure for defective parts.

3.3.5 Fastening devices.- All screws, pins, and bolts shall be installed with means to prevent loosening. All parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.6 Threads.- All threaded parts shall be in the inch system and shall conform to Handbook H-28.

3.3.7 Bearing lubrication.- All bearings (except sealed for life type) and other moving parts shall have accessible means for lubrication.

3.3.8 Interchangeability.- All parts shall be manufactured to dimensional tolerances that shall permit replacement without modification.

3.4 Riveters.-

* 3.4.1 Type I riveters.- The type I riveters shall have alligator jaws and shall be similar in configuration to figure 1. Dimensions, reach capacity, force capacity, and weight shall conform to the respective requirements specified in table I. The jaws shall be made of alloy steel, heat treated to withstand the forces specified in table I for each respective class (see 4.3.2). One jaw shall be stationary and the other jaw moveable. The jaws shall accommodate and retain the rivet sets during the riveting operation and permit hand removal and insertion of rivet sets. The type I riveters shall be tested as specified in 4.3.3 for force as specified in table I.

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* 3.4.2 Type II riveters.- The type II riveters shall be the "C" yoke type, similar in configuration to figure 2. Dimensions, reach capacity, force capacity, and weight shall conform to table II. The jaws shall be heat treated to withstand the forces specified in table II (see 4.3.2). The "C" yoke shall have a stationary jaw and a moving plunger. The jaws shall accommodate and retain the rivet sets during the riveting operation and permit hand removal and insertion of rivet sets. When specified (see 6.2), the riveter shall be provided with an automatic plunger return. The type II riveters shall be tested as specified in 4.3.3 for force as specified in table II.

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TABLE I.- Type I-Alligator-jaw compression riveter

Class	Capacity cold rivet Diameter		Yoke dimensions		Moving jaw Rivet set travel			Riveter	
Class	Alum. Alloy (Inch)	Steel (Inch)	Reach capacity (minimum) (Inches)	Closed height 1/ 11/16 -0 (Inch)	Shall equal or exceed travel (Inches)	Shall equal or exceed force with 90 lb. air pressure (Pounds)	Part of stroke at maximum force (Inch)	Max. weight (Pounds)	Max. overall length (Inches)
1	1/3	3/32	1-3/8	7/8	1/2	3,000	Final 1/16	4-1/2	10-3/16
1L	1/3	3/32	9-1/8	3/4	1-11/16	3,000	Final 1/16	26-1/2	26-1/2
2	5/32	1/8	5	7/8	1-3/8	3,400	Final 3/16	20-1/4	22-1/2
3	3/16	5/32	2-7/8	7/8	5/8	5,200	Final 1/8	16-1/4	25

1/ Dimensions between jaws when the riveter jaws are in a fully closed position with both anvils or rivet sets removed.

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TABLE II.- Type II-C-yoke compression riveter

Class	Capacity cold rivet Diameter		Yoke dimensions		Rivet set travel			Riveter	
Class	Alum. Alloy (Inch)	Steel (Inch)	Reach capacity (minimum) (Inches)	Closed height 1/ +1/16 -0 (Inch)	Shall equal or exceed travel (Inches)	Shall equal or exceed force with 90 lb. air pressure (Pounds)	Part of stroke at maximum force (Inch)	Max. weight (Pounds)	Max. overall length (Inches)
1	1/8	3/32	1-3/8	11/16	1/2	3,000	Final 1/16	5-1/4	10-3/4
3	3/16	5/32	2	25/32	1/2	6,000	Final 1/8	12-3/4	18-1/8

1/ Dimensions between jaws when the riveter jaws are in a fully closed position with both anvils or rivet sets removed.

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* 3.5 Rivet sets.- Unless otherwise specified (see 6.2), a group of rivet sets shall be supplied with each riveter. The group shall consist of cupped and flush rivet sets to drive universal head rivets conforming to MS 20470, MS 20613 and MS 20615, having shank size, 1/8 inch, 5/32 inch, and 3/16 inch diameter. Table III specifies the rivet sets to be included for each type and class of riveter. The design and dimensions shall conform to figures 3 and 4. The rivet set shall be made of steel and shall be heat treated to withstand the functional test specified in 4.3.2. A noncorrosive container with a cover and latch shall be supplied with each group of rivet sets. The container shall be constructed of metal or plastic. The container shall have individual compartments for holding each rivet set.

* TABLE III.- Flush and cupped-head rivet set selection chart
(see figures 3 and 4 "A" dimensions)

Rivet length (inch)	Type I, classes 1, 2, and 3 Type II, class 3 Rivet diameter (inch)			Type I, class 1L Rivet diameter (inch)			Type II, class 1 Rivet diameter (inch)		
	1/8	5/32	3/16	1/8	5/32	3/16	1/8	5/32	3/16
	Rivet-set head "A" lengths (see figures 3 and 4)								
3/8	3/16	3/16	1/4	3/16	3/16	3/16	3/16	3/16	3/16
	1/2	35/64	17/32	3/8	27/64	15/32	5/16	23/64	13/32
5/16	3/16	3/16		3/16	3/16		3/16	3/16	
	9/16	39/64		7/16	31/64		3/8	27/64	
1/4	3/16			3/16	3/16		3/16		
	5/8			1/2	35/64		1/2		

* 3.6 Finish.- All exposed ferrous surfaces shall be painted or plated in accordance with the manufacturer's commercial practice, except that cadmium plating shall not be allowed.

3.7 Marking.- Each riveter and each rivet set, when applicable, shall be permanently marked with the manufacturer's model, type, or size designation, together with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

3.8 Workmanship.- Riveters shall be free of scale, mismatching, peening, plugging, or filling with cold solders, and metallic pastes. The riveters shall be free of sand, grit, fins, corrosion, sprues, flux. External surfaces shall be smooth and all edges rounded or beveled. All fasteners shall be tight and all components in alignment.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 End item inspection.- A lot shall consist of all riveters of one type and class offered for inspection at one time. The sample unit shall be one completely fabricated riveter, and rivet sets, when applicable.

4.2.2.1 Visual examination.- The end item shall be examined for the defects classified in table IV. The inspection level shall be level II and the acceptable quality level (AQL) shall be 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE IV.- Classification

Examine	Defect	Classification Major	Minor
Finish	Not manufacturer's commercial plated or painted finish		X
Workmanship and construction	External surfaces not smooth, edges not rounded or beveled		X
	Surfaces not free of scale, mismatching, peening, plugging, filling with cold solder or metallic pastes		X
	Surfaces not free of sand, grit, fins, corrosion, sprues, and flux		X
	Fasteners not tight (i.e. can be loosened by hand)	X	
	Components not in alignment	X	

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TABLE IV.- Classification (cont'd)

Examine	Defect	Classification Major	Minor
Workmanship and construction (cont'd)	Air strainer missing		X
	Lubricating means missing, when applicable		X
	Rivet sets missing, when applicable		X
	Rivet sets not size required, when applicable		X
	Container missing when applicable		X
	Container not provided with individual compartments		X
Marking	Missing, incomplete, illegible		X

4.2.2.2 Dimensional examination.- Examination shall be made of the end item for compliance with dimensions and weights specified. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.2.3 Testing of the end item.- Riveters shall be tested in accordance with 4.3.1 and 4.3.2. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.4 Examination of preparation for delivery.- Examination of preparation for delivery shall be in accordance with PPP-P-40.

4.3 Tests.-

4.3.1 Operational test.- Each riveter shall be connected to an air supply source delivering not less than 90 psig. The throttle shall be gradually depressed in a manner to permit the riveter to go from a static position to full speed. Full speed shall be maintained for a minimum of 5 minutes. After shutting off the air supply, the riveter shall be examined for compliance with 3.4. Any failure of operation, fracture, deformation due to vibration, or loosening of fasteners shall constitute failure of this test.

* 4.3.2 Functional test.- Operating from a 90 psig air supply source, the riveter shall cold drive 25 rivets each conforming to MS 20470, MS 20613 and MS 20615, and tightly rivet two pieces of sheet metal. Rivets, both 1/8 and 3/16 inch diameter, in the shortest and longest lengths, shall be tested with two pieces of sheet metal. After cutting off the air supply, the riveter and rivet sets shall be examined for compliance with 3.4.1, 3.4.2 and 3.5. Any damage to the riveter, jaws or rivet sets shall constitute failure of this test.

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4.3.3 Pressure pounds test.- Each riveter shall be tested with 90 pounds air pressure as specified in 3.4.1 and 3.4.2 to determine compliance with tables I and II.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking shall be in accordance with PPP-P-40 for grinder, pneumatic, portable. Level of preservation and packaging and level of packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use.- The riveters covered by this specification are intended for general utility maintenance usage to perform various riveting operations in conjunction with sheet-metal bench work, modifications, and installations.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Type and class required (see 1.2)
(c) When automatic plunger return is required (see 3.4.2)
(d) When rivet sets and rivet set containers are not required (see 3.5)
(e) Selection of applicable levels of preservation, packaging and packing (see 5.1)

6.3 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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FIG. 1 - ALLIGATOR SQUEEZE RIVETER

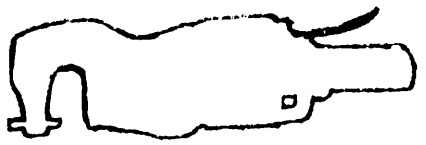
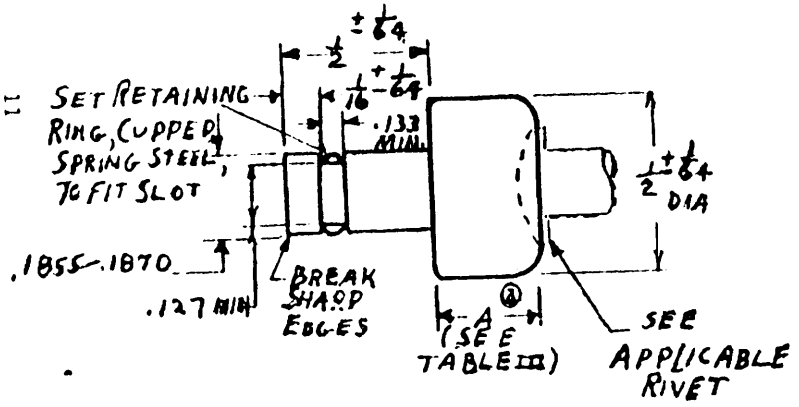


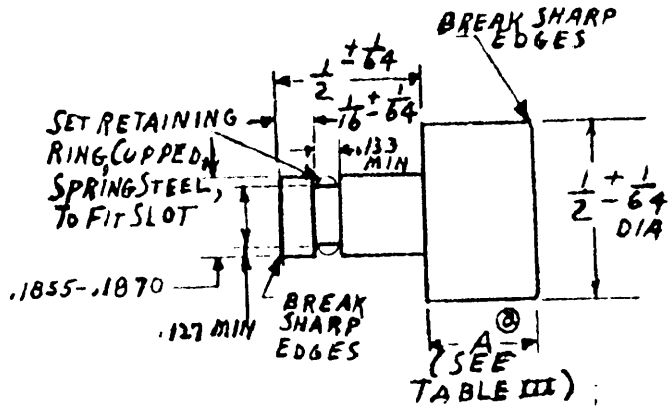
FIG 2. - SQUEEZE RIVETER WITH "C" YOKE



@ LENGTH ± .020 INCH

DIMENSIONS IN INCHES

FIG. 3. CUPPED SET DESIGN



@ LENGTH ± .020 INCH

DIMENSIONS IN INCHES

FIG. 4. FLUSH OR FLAT SET

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YOKE TERMINOLOGY

Rivet Set — Term generally used to describe any one rivet die.

Dolly — Term frequently used to describe the rivet set which is mounted in the stationary set holder.

IMPORTANT

Rivet sets are not in place in the illustration below.

X — Extreme open position
Y — Extreme closed position

A — Reach
B — Closed height
C — Throat gap
D — Travel
E — Total yoke gap
F — Anvil work clearance
G — Upper offset
H — Lower offset

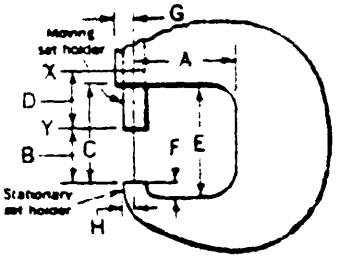


FIGURE 5 YOKE TERMINOLOGY

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

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b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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